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Framing Spatial Hearing within the Free Energy Principle: A Hierarchical Model of Perceptual Learning

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Abstract. Spatial hearing relies on dynamic internal representations to decode spatial information from ever-changing acoustic conditions. A fundamental example is sound localisation, where listeners infer the direction of an invisible sound. Bayesian inference provides a computational account of such a localisation process; however, modelling efforts are largely limited to static listening scenarios. Instead, the free energy principle (FEP) describes perception, learning, and action as joint minimisation of variational free energy (VFE), providing a principled framework to account for dynamic conditions. Here, we take a first step toward applying the FEP to spatial hearing. We characterise how listeners' beliefs evolve across trials, in which they estimated sound direction and received subsequent visual feedback on each trial. Listeners are described by a hierarchical generative model where visual feedback drives gradient descent on VFE, updating beliefs over three independent quantities: lateral and polar localisation precisions and hemifield disambiguation. The model captures individual learning trajectories and reveals components with different dynamics: lateral precision improves, as does hemifield disambiguation, consistent with increased sensitivity of the auditory pathway in decoding vertical direction. Notably, polar precision also increases, most for listeners with the lowest initial precision, suggesting that the observed vertical improvement is driven by both hemifield disambiguation and polar precision. This work establishes spatial hearing as a tractable domain for FEP and motivates future closed-loop experiments with full action-perception cycles.

Introduction. Understanding how the auditory system decodes spatial information is a central challenge in hearing research, yet computational accounts largely focus on static scenarios [12, 4], leaving its dynamic, experience-dependent nature under-explored [3]. Auditory spatial perception is inherently plastic [9] and localisation involves both perception and action [14], making the FEP, a unified computational account of perception, learning, and action [8, 16], a natural framework. Localisation improves across trials with visual feedback [11], yet no mechanistic account exists of how individual beliefs evolve trial by trial. Here, we use the FEP [5] to model perceptual learning as belief revision, decomposing lo-

calisation improvement into three components with distinct timescales, validated on behavioural data from [11].

Problem formulation. On each trial i, the listener estimates the direction of an invisible sound source $\boldsymbol{\varphi}_t$ and produces a pointing response $\boldsymbol{\varphi}_r$, followed by visual feedback at $\boldsymbol{\varphi}_t$. Directions are expressed in interaural-polar coordinates $\boldsymbol{\varphi} = (\alpha, \beta)$ [15], separating the contributions of auditory cues to behavioural error (Fig. 1): the lateral angle α, informed by interaural time and level differences, defines the cone of confusion, within which the polar angle β is decoded from spectral-shape cues [14]. With uniform priors on $\boldsymbol{\varphi}_r$ and $\boldsymbol{\varphi}_t$, the listener model factorises as $p(\boldsymbol{\varphi}_r|\boldsymbol{\varphi}_t, \boldsymbol{\theta}) = p(\alpha_r|\alpha_t, \boldsymbol{\theta})\, p(\beta_r|\alpha_r, \beta_t, \boldsymbol{\theta})$ where $\boldsymbol{\theta}$ captures the listener's beliefs about lateral and polar precisions, and hemifield disambiguation (i.e., front–back confusions [14]).

We employ a parametric formulation to preserve differentiability within FEP. The lateral component is a von Mises distribution with location α_t and concentration κ_L (i.e., lateral precision). The polar component is a two-component von Mises mixture with concentration κ_P (i.e., polar precision), locations β_t and $\pi - \beta_t$ (i.e., correct and incorrect hemifields) and hemifield weight w; κ_P is scaled by $\cos^2(\alpha_r)$, reflecting the reduced relevance of spectral cues at lateral extremes, where the interaural cone collapses (Fig. 1) [15, 13].

Belief revision follows from minimising variational free energy $\mathcal{F}$ with respect to $\boldsymbol{\theta} = (\kappa_L, \kappa_P, w)$. With a flat prior and Laplace approximation [6], $\mathcal{F}$ reduces to $-\log p(\boldsymbol{\varphi}_r|\boldsymbol{\varphi}_t, \boldsymbol{\theta})$, so belief revision amounts to online likelihood ascent: a deliberately minimal instance of the FEP, covering perception and learning but not action (Fig. 2). Each belief takes a natural-gradient step [2] after every trial, scaled by the inverse Fisher information to update components living on different scales consistently (problem formulation available in Appendix A.1). Updates are sequential, producing a trajectory that batch optimisation cannot recover. Per subject (N = 5), learning rates (η_L, η_P, η_w) were estimated by maximum likelihood [1], and $\boldsymbol{\theta}_0$ from the baseline block [11] (Appendix A.2).

Results and Discussion. The model captures group-average trajectories ($R^2 = 0.75, 0.83, 0.79$ for lateral, polar and hemifield errors, Fig. 3, top). Individual parameter trajectories (Fig. 3, bottom) differ across components: κ_L rises for most subjects; κ_P rises, proportionally more for lower baseline κ_P; w increases steadily (median $\eta_L, \eta_P, \eta_w = 0.010, 0.015, 0.014$; Appendix A.3). These distinct trajectories are consistent with previously reported mechanisms, including sensorimotor recalibration [11], stable spectral cue-to-direction mappings acquired through natural listening [7], and increased sensitivity to altered spectral cues [10]. Notably, behavioural polar dispersion decreases and κ_P rises under the $\cos^2$ weighting: the improvement seen in aggregate statistics stems from hemifield disambiguation and sharper polar precision (Fig. 3, middle column).

Our approach has limitations: a small cohort and a perception-only loop in which pointing is a read-out and feedback is external. Two extensions would cast spatial hearing as full active inference: informative priors on $\boldsymbol{\theta}$, yielding posterior uncertainty on κ_P, and head movements as policies, resolving hemifield ambiguity by action rather than by learning.

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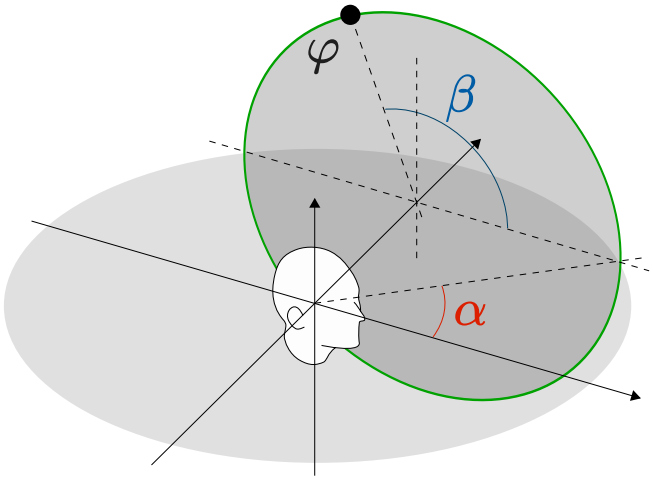


Fig. 1. Interaural-polar coordinate system [15]. The source direction is defined by $\boldsymbol{\varphi} = (\alpha, \beta)$: lateral angle α identifies the interaural axis and the cone of confusion (green circle); polar angle β defines elevation within that cone. Reference points: front (0°, 0°), back (0°, 180°), sides (±90°, 0°).

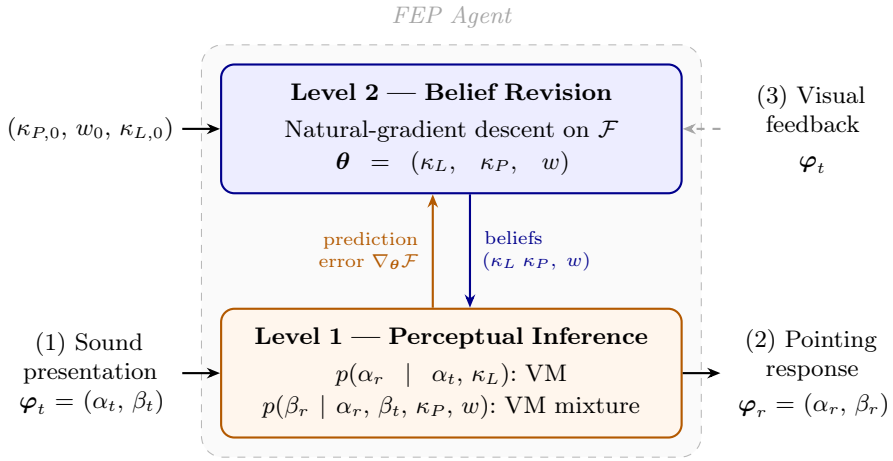


Fig. 2. Architecture of the hierarchical FEP agent. Level 1 encodes the listener's generative model of auditory space as a von Mises mixture (polar) and a single von Mises (lateral). Level 2 revises the agent's beliefs $\boldsymbol{\theta} = (\kappa_L, \kappa_P, w)$ via natural-gradient descent on variational free energy $\mathcal{F}$, using visual feedback $\boldsymbol{\varphi}_t = (\alpha_t, \beta_t)$ as a high-precision teaching signal. Initial beliefs are estimated from baseline block available from the original experiment (naive localisation without feedback [11]) by maximum likelihood. Learning rates (η_L, η_P, η_w) are fitted per subject.

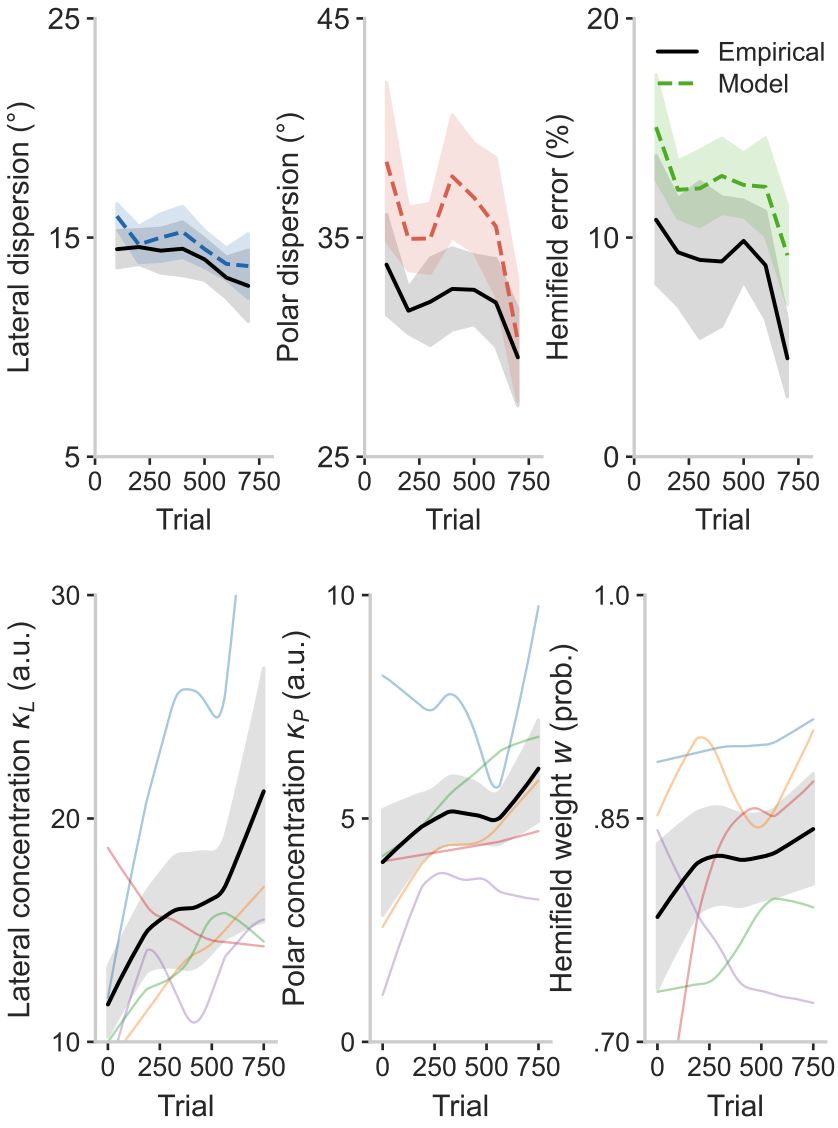


Fig. 3. *Top:* Group-average behavioural trajectories, manual condition (n=5). Solid: empirical (mean ± 1 standard error (SE) ribbon); dashed: FEP model. Left to right: lateral dispersion (standard deviation (SD) of lateral error, °); polar dispersion (SD of polar error excluding hemifield errors, $\cos^2 \alpha_t$-weighted, °); hemifield error rate (proportion of responses deviating more than 90° from the target, $\cos^2 \alpha_t$-weighted, %). The model captures the direction and shape of all three metrics. *Bottom:* Individual parameter trajectories, manual group (n=5), truncated to 750 trials (LOESS-smoothed, span = 0.5; thin coloured lines: individual subjects; thick black line: mean ± 1 SE ribbon).

A Appendix

Code and data are available here:
`https://github.com/robaru/iwai2026-fep-spatial-hearing`.

A.1 Model formulation

Generative model Directions are expressed in interaural-polar coordinates $\boldsymbol{\varphi} = (\alpha, \beta)$ [15]. Circular quantities are modelled with the von Mises distribution

$$\mathcal{VM}(x;\, \mu, \kappa) = \frac{e^{\kappa \cos(x-\mu)}}{2\pi I_0(\kappa)}, \tag{A1}$$

where I_0 is the modified Bessel function of order zero and κ the concentration (precision). The listener model factorises into a lateral and a polar component, $p(\boldsymbol{\varphi}_r|\boldsymbol{\varphi}_t, \boldsymbol{\theta}) = p(\alpha_r|\alpha_t, \kappa_L)\, p(\beta_r|\alpha_r, \beta_t, \kappa_P, w)$, with

$$p(\alpha_r|\alpha_t, \kappa_L) = \mathcal{VM}(\alpha_r;\, \alpha_t, \kappa_L), \tag{A2}$$

$$p(\beta_r|\alpha_r, \beta_t, \kappa_P, w) = w\, \mathcal{VM}(\beta_r;\, \beta_t, \tilde{\kappa}_P) + (1-w)\, \mathcal{VM}(\beta_r;\, \pi - \beta_t, \tilde{\kappa}_P), \tag{A3}$$

where $\tilde{\kappa}_P = \kappa_P \cos^2(\alpha_r)$ accounts for the geometric compression of the cone of confusion at lateral angles, and $w \in (0, 1)$ is the probability of resolving the correct hemifield. The two mixture components share the same concentration: hemifield confusion affects where the response lands, not how tightly it clusters.

Free energy and natural gradient For a recognition density $q(\boldsymbol{\theta})$ over the beliefs $\boldsymbol{\theta} = (\kappa_L, \kappa_P, w)$, the variational free energy at trial i is

$$\mathcal{F}_i = \mathrm{KL}[q(\boldsymbol{\theta}) \,\|\, p(\boldsymbol{\theta})] - \mathbb{E}_q[\log p(\boldsymbol{\varphi}_r|\boldsymbol{\varphi}_t, \boldsymbol{\theta})]\,. \tag{A4}$$

Under a Laplace approximation (q Gaussian centred on the current point estimate $\boldsymbol{\theta}_i$) and a flat prior, the complexity term is constant in $\boldsymbol{\theta}_i$, so $\mathcal{F}_i = -\log p(\boldsymbol{\varphi}_r|\boldsymbol{\varphi}_t, \boldsymbol{\theta}_i) + \text{const}$ [6]. Beliefs are revised by natural-gradient descent [2],

$$\boldsymbol{\theta}_{i+1} = \boldsymbol{\theta}_i - \boldsymbol{\eta}\, \mathcal{I}(\boldsymbol{\theta}_i)^{-1} \nabla_{\boldsymbol{\theta}} \mathcal{F}_i, \tag{A5}$$

where $\boldsymbol{\eta} = \mathrm{diag}(\eta_L, \eta_P, \eta_w)$ collects the learning rates and $\mathcal{I}$ is the Fisher information. For a von Mises concentration, $\mathcal{I}(\kappa) = A'(\kappa) = 1 - A(\kappa)^2 - A(\kappa)/\kappa$, with $A(\kappa) = I_1(\kappa)/I_0(\kappa)$ the Bessel ratio, i.e., the expected value of $\cos(x-\mu)$. For the hemifield weight, parameterised through its logit $\psi = \log \frac{w}{1-w}$, $\mathcal{I}(\psi) = w(1-w)$; in the update, $\boldsymbol{\theta}$ therefore denotes $(\kappa_L, \kappa_P, \psi)$, with w recovered through the logistic function, $w = \sigma(\psi)$. Because the likelihood factorises, the three updates are decoupled.

Update equations Let $\mathcal{VM}_c = \mathcal{VM}(\beta_r;\, \beta_t, \tilde{\kappa}_{P,i})$ and $\mathcal{VM}_m = \mathcal{VM}(\beta_r;\, \pi - \beta_t, \tilde{\kappa}_{P,i})$ denote the correct- and mirror-hemifield components of the polar mixture evaluated at the response on trial i, with $\tilde{\kappa}_{P,i} = \kappa_{P,i}\cos^2(\alpha_t)$, and $p_i = w_i\mathcal{VM}_c + (1 - w_i)\mathcal{VM}_m$ the polar mixture likelihood. The updates are

$$\kappa_{P,i+1} = \kappa_{P,i} + \frac{\eta_P \cos^2(\alpha_t)}{A'(\kappa_{P,i})}\big[r_i - A(\tilde{\kappa}_{P,i})\big], \tag{A6}$$

$$\psi_{i+1} = \psi_i + \eta_w \frac{\mathcal{VM}_c - \mathcal{VM}_m}{p_i}, \qquad w_{i+1} = \sigma(\psi_{i+1}), \tag{A7}$$

$$\kappa_{L,i+1} = \kappa_{L,i} + \frac{\eta_L}{A'(\kappa_{L,i})}\big[\cos(\alpha_r - \alpha_t) - A(\kappa_{L,i})\big], \tag{A8}$$

where σ is the logistic function and

$$r_i = \frac{w_i \cos(\beta_r - \beta_t)\,\mathcal{VM}_c + (1 - w_i)\cos(\beta_r - \pi + \beta_t)\,\mathcal{VM}_m}{p_i} \tag{A9}$$

is the responsibility-weighted cosine alignment between response and target. Each bracketed term is a prediction error: the observed alignment minus the alignment expected under the current belief, for κ_P evaluated at the scaled concentration $\tilde{\kappa}_{P,i}$. The cone of confusion enters twice. At perception time the listener has no access to the true lateral angle, so the generative model above conditions on the inferred cone α_r. At update time the visual feedback reveals the true direction, so $\mathcal{F}$ in Eq. A4 is evaluated with the true cone and the κ_P step in Eq. A6 is scaled by $\cos^2(\alpha_t)$, the same factor used in the likelihood that fits $\boldsymbol{\theta}_0$ and the learning rates. For w, the Fisher information cancels the logistic derivative exactly, and the logit parameterisation keeps $w \in (0, 1)$ without projection.

A.2 Parameter estimation

Initial beliefs $\boldsymbol{\theta}_0$ were estimated per subject by maximum likelihood from the last 100 trials of the baseline block (naive localisation without feedback [11]). The learning rates (η_L, η_P, η_w) are the only free parameters of the learning model. They were fitted per subject by minimising the mean negative log-likelihood of the responses in the feedback block under the sequential updates of Eqs. A6–A8, using Bayesian adaptive direct search [1]. Because the likelihood factorises, (η_P, η_w) and η_L are estimated independently.

A.3 Model comparison and robustness

Held-out comparison. To test whether all three learning rates are needed, they were fitted on the odd-numbered feedback trials and the summed log-likelihood was evaluated on the even-numbered trials, with beliefs updated sequentially over all trials in both cases (Table 1, left). Relative to the full model, freezing κ_P ($\eta_P = 0$) reduced the held-out log-likelihood by 167 nats in total and freezing w ($\eta_w = 0$) by 82 nats. Hemifield learning was required in every subject (NH12

only marginally), and polar-precision learning in all but NH17, for which η_P was indistinguishable from zero (parametric bootstrap, $B = 100$, 95% interval reaching zero). Over the feedback block, κ_P rose by 2.8 to 4.5 in the other four subjects (95% intervals excluding zero) and by 0.3 in NH17, with the largest relative gain in the two subjects with the lowest baseline κ_P (NH15, NH18). A first-half/second-half split gave the same ordering.

Robustness to the $\cos^2$ *scaling.* The Laplace approximation with a flat prior affects only the curvature of $\mathcal{F}$, not the location of the update, so it cannot mask a change in κ_P. The $\cos^2(\alpha)$ scaling changes the size of polar learning but not its direction (Table 1, right): removing it from likelihood, update and baseline fit lowers the median η_P from 0.015 to 0.006, κ_P still rises by 0.6 to 2.8 over the feedback block, and the unscaled model fits worse with the same number of parameters (summed in-sample $\Delta\mathrm{LL} = -53$ nats, although NH15 and NH17 favour it). The rise of w and the necessity of hemifield learning are unaffected. The rise of κ_P is therefore not conditional on the geometric weighting of lateral targets. The unscaled model, however, does not respect the problem geometry: the cone of confusion collapses to a point at lateral positions ($\alpha = \pm 90°$), where the polar angle is undefined, and polar errors become progressively less informative about spectral decoding as $|\alpha|$ increases.

Table 1. Left: held-out comparison, log-likelihood differences in nats relative to the full model ($\Delta\mathrm{LL} < 0$: reduced model predicts worse). Right: $\cos^2$ robustness, polar-precision learning rate η_P with bootstrap 95% interval, with and without the $\cos^2(\alpha)$ scaling, and $\Delta\kappa_P$, the change of κ_P over the feedback block without scaling.

		Held-out comparison		$\cos^2$ robustness		
Subject	$\kappa_{P,0}$	$\Delta\mathrm{LL}_{\eta_P=0}$	$\Delta\mathrm{LL}_{\eta_w=0}$	η_P ($\cos^2$)	η_P (no $\cos^2$)	$\Delta\kappa_P$
NH12	7.5	−5.7	−1.1	0.020 [0.015, 0.033]	0.005 [0.002, 0.010]	+1.9
NH15	2.4	−54.2	−6.5	0.015 [0.013, 0.026]	0.016 [0.013, 0.025]	+2.8
NH16	4.3	−39.0	−39.0	0.008 [0.006, 0.012]	0.006 [0.005, 0.010]	+1.6
NH17	4.3	0.0	−30.2	0.002 [0, 0.012]	0.006 [0, 0.016]	+0.6
NH18	0.9	−68.2	−5.0	0.062 [0.038, 0.084]	0.043 [0.034, 0.079]	+2.4